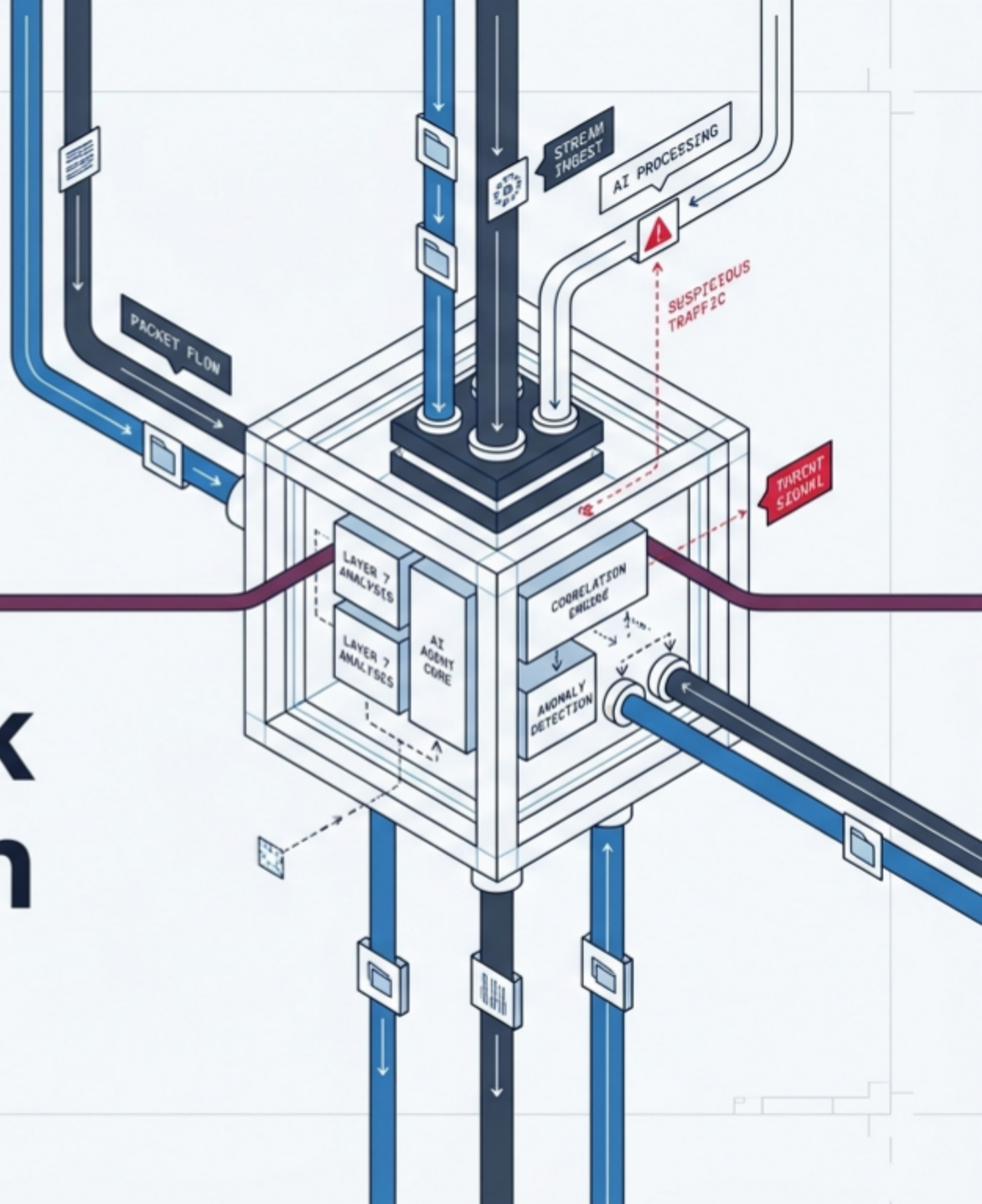
FROM RAW PACKET TO HUMAN-READABLE INTELLIGENCE

Real-Time Network Intrusion Detection

An Agentic AI-Enhanced Architecture for Real-Time Detection, Correlation, and Automated Reporting.



The Problem: Alert Fatigue

Modern NIDS effectively detect threats, but security teams are overwhelmed by isolated alerts lacking historical context, incident-level grouping, and clear explanations.



The Solution: Contextual AI

A hybrid NIDS augmented with an Agentic AI layer to automatically correlate, contextualize, prioritize, and explain incidents in real-time, bridging the gap between raw data and analyst action.

Data Ribbon

INCIDENT CARD: ID #2024-AI-001

STATUS: RESOLVED (AI-ASSISTED)

SEVERITY: CRITICAL (CRIMSON BADGE)

TIME WINDOW: 10:45 AM - 11:02 AM

CORRELATED ALERTS:

- ID 10234: PORT SCAN (443/TCP)
- ID 10235: SQL INJECTION ATTEMPT
- ID 10236: MALWARE SIGNATURE (MIRAI BOTNET)
- ID 10237: LATERAL MOVEMENT (SMB)

AI CONTEXTUALIZATION:

AGENTIC AI CORE (DEEP PLUM ICON): A coordinated multi-stage attack initiated with a port scan followed by an SQL injection. The attacker gained entry, deployed a Mirai botnet variant, and attempted lateral movement via SMB.

PRIORITY SCORE: 98/100 (CRIMSON BADGE)

EXPLANATION: High likelihood of successful data breach and network compromise. Immediate action required.

RECOMMENDED ACTION: Isolate affected host (IP 182.168.1.100), block attacker IP (203.0.113.55), patch SQL vulnerability, review access logs.

EXECUTE RESPONSE

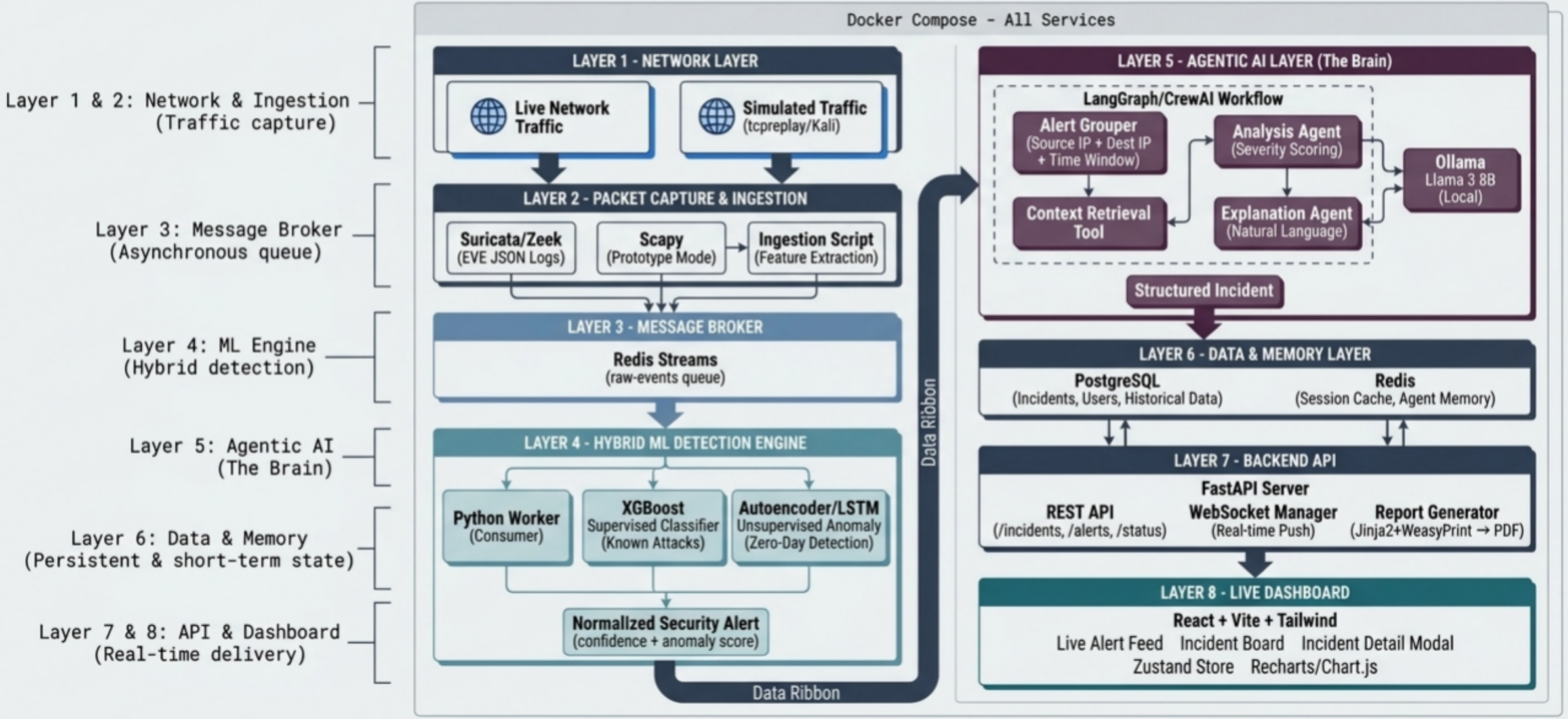
VIEW FULL REPORT

REAL-TIME STREAMING DATA FLOW

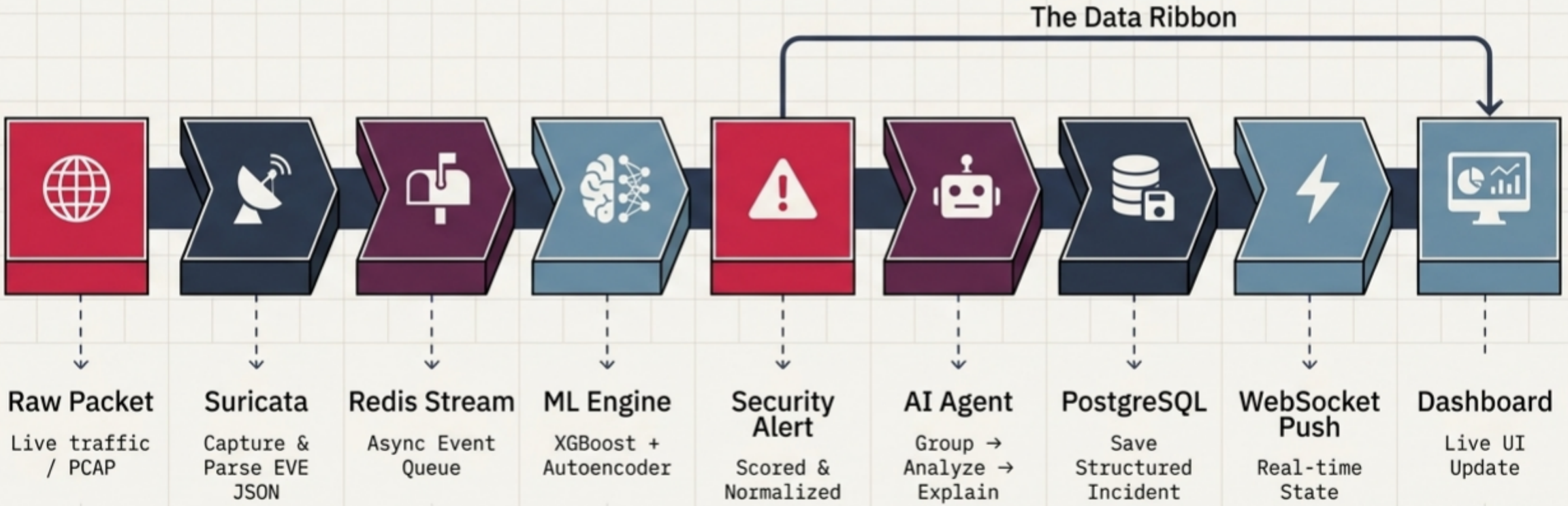
Editorial Technical Blueprint

End-to-End System Architecture

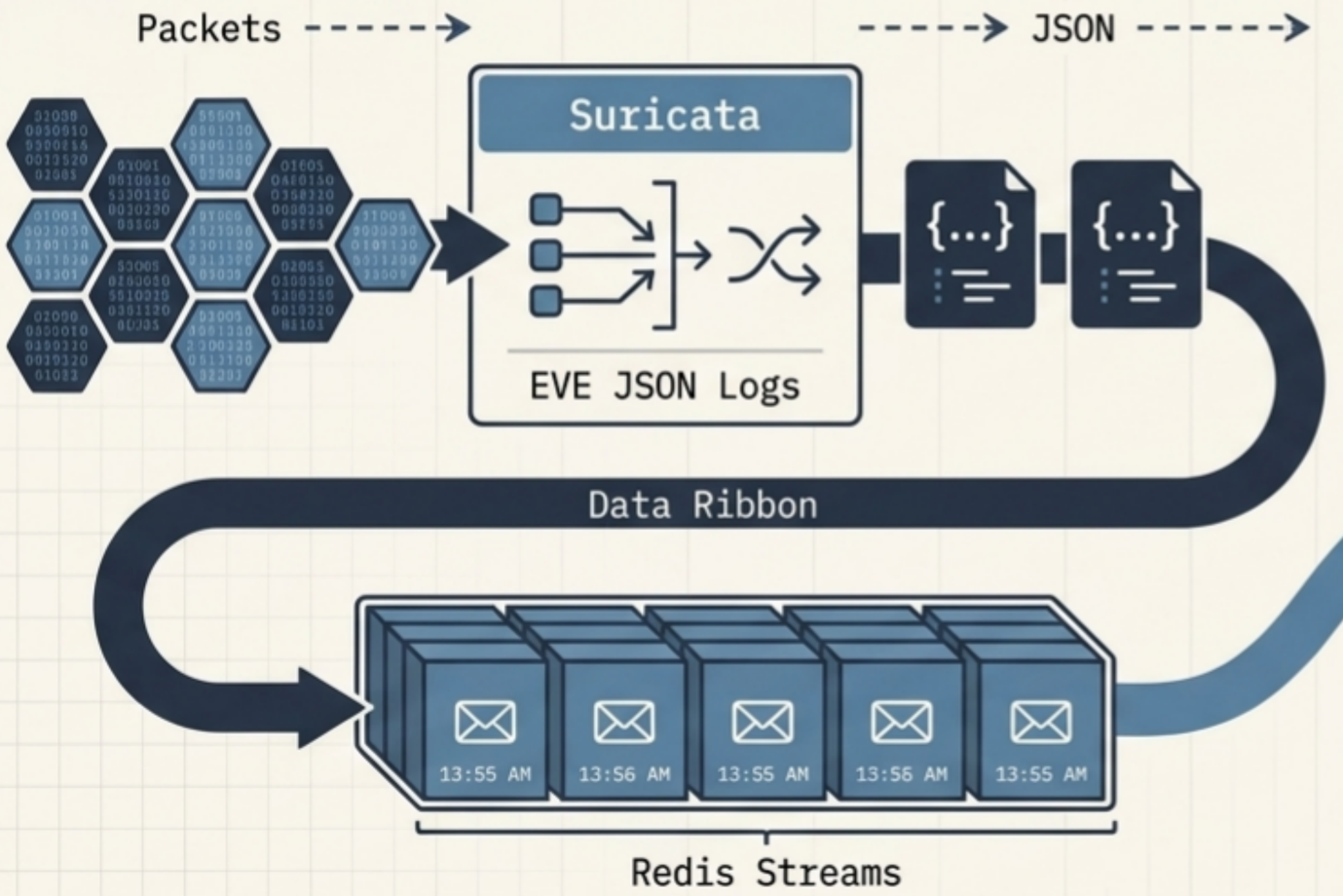
Note: The AI is NOT used for raw packet classification. It is a contextual, tool-assisted incident analyst.



Packet to Incident:
The 9-Step Data Pipeline



Layer 1-3: Capture, Ingestion & Queuing



The Parser: Suricata/Zeek

Replaces prototype tools (Scapy). Industry standard capable of 10Gbps+ parsing, outputting structured flow and protocol features via EVE JSON directly into the ML pipeline.

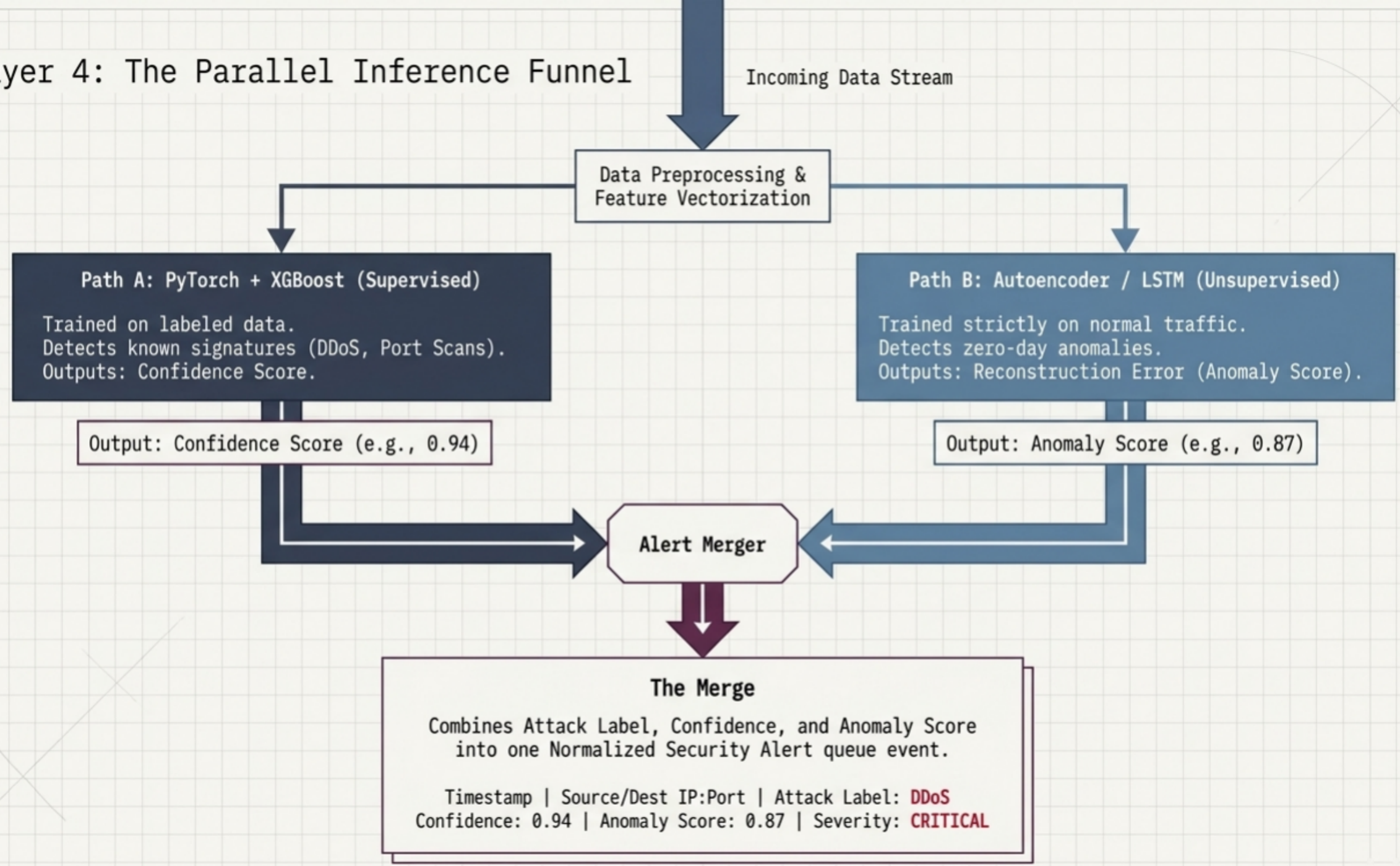
The Shock-Absorber: Redis Streams

Introduces a message broker layer. Decouples high-speed packet capture from the heavier ML engine and Agentic layers.

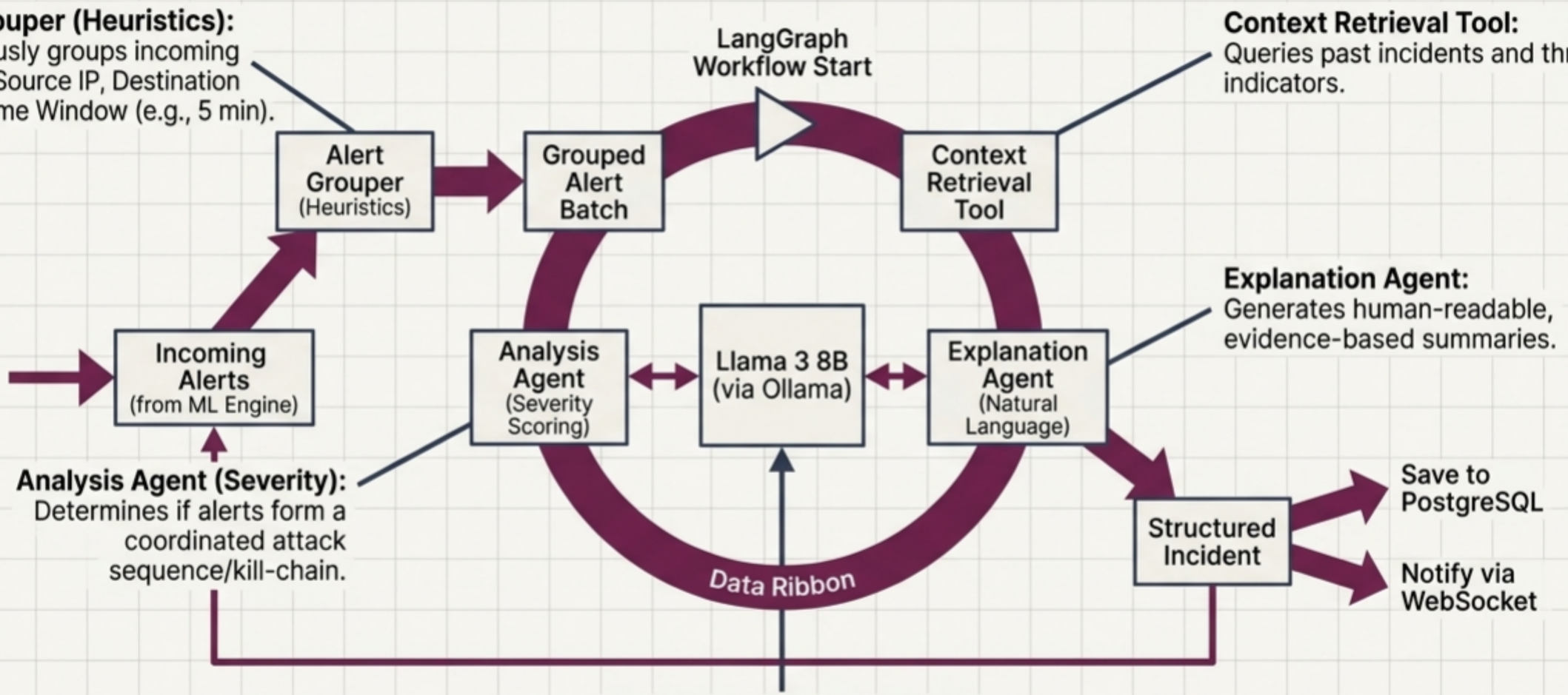
Architectural Rationale

Guarantees zero dropped packets during massive network traffic spikes or DDoS events.

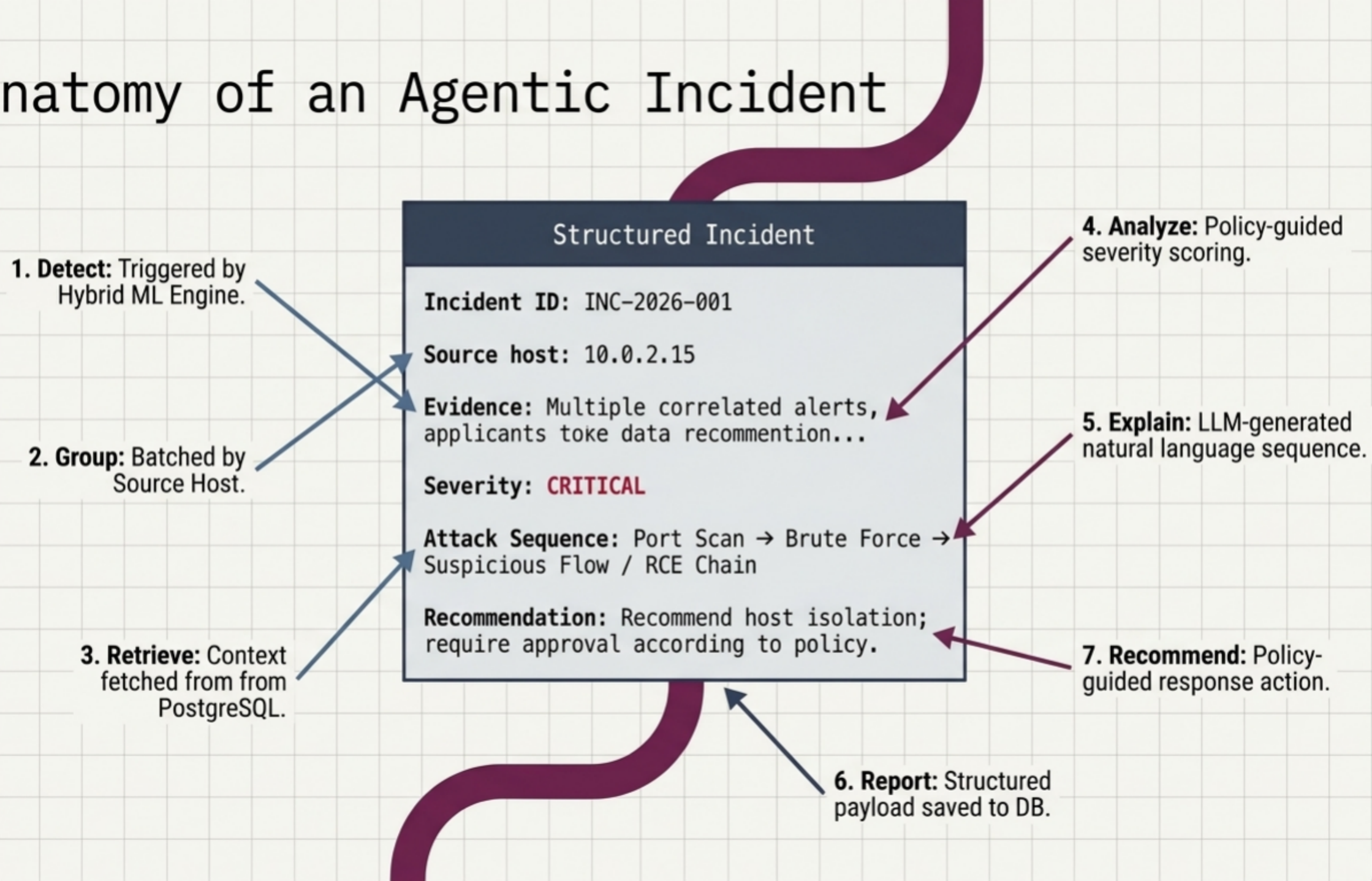
Layer 4: The Parallel Inference Funnel



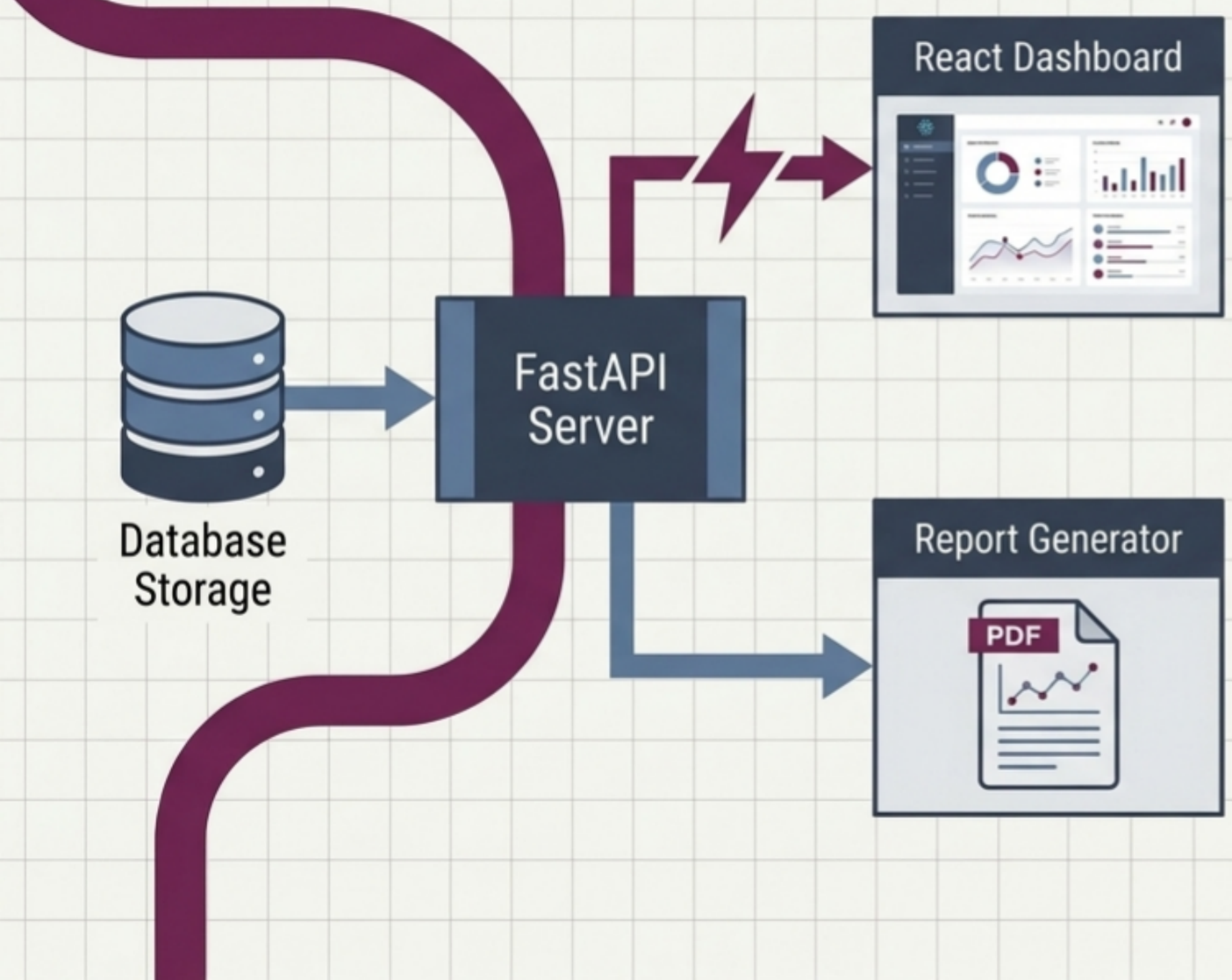
Layer 5: The Agentic Memory Loop



Anatomy of an Agentic Incident



Layers 6-8: Real-Time State & Delivery



Backend & APIs:

FastAPI manages standard REST CRUD operations and asynchronous **WebSocket** streaming.

Real-Time State:

WebSockets push new incidents instantly. The `React/Vite` frontend uses `Zustand` to manage complex, rapidly changing UI state without page reloads.

Automated Reporting:

An endpoint triggers `Jinja2` and `WeasyPrint` to instantly render HTML templates of incidents into downloadable PDF reports.

Tech Stack Evaluation: Prototype to Production

Component	Prototype Standard	Refined Production Stack	Architectural Rationale
Packet Capture	Scapy	Suricata/Zeek	Handles 10Gbps+ without dropping packets.
Message Broker	None	Redis Streams	Decouples capture from ML to prevent traffic spike failures.
ML Pipeline	Scikit-learn	PyTorch + XGBoost	Industry standard for tabular flow data and deep learning.
Agentic AI	Basic LLM	LangGraph + Llama 3 8B	Advanced multi-step tool use, local/private execution.

Deployment Topology (Docker Orchestration)

Row 3: Application Layer (Top)

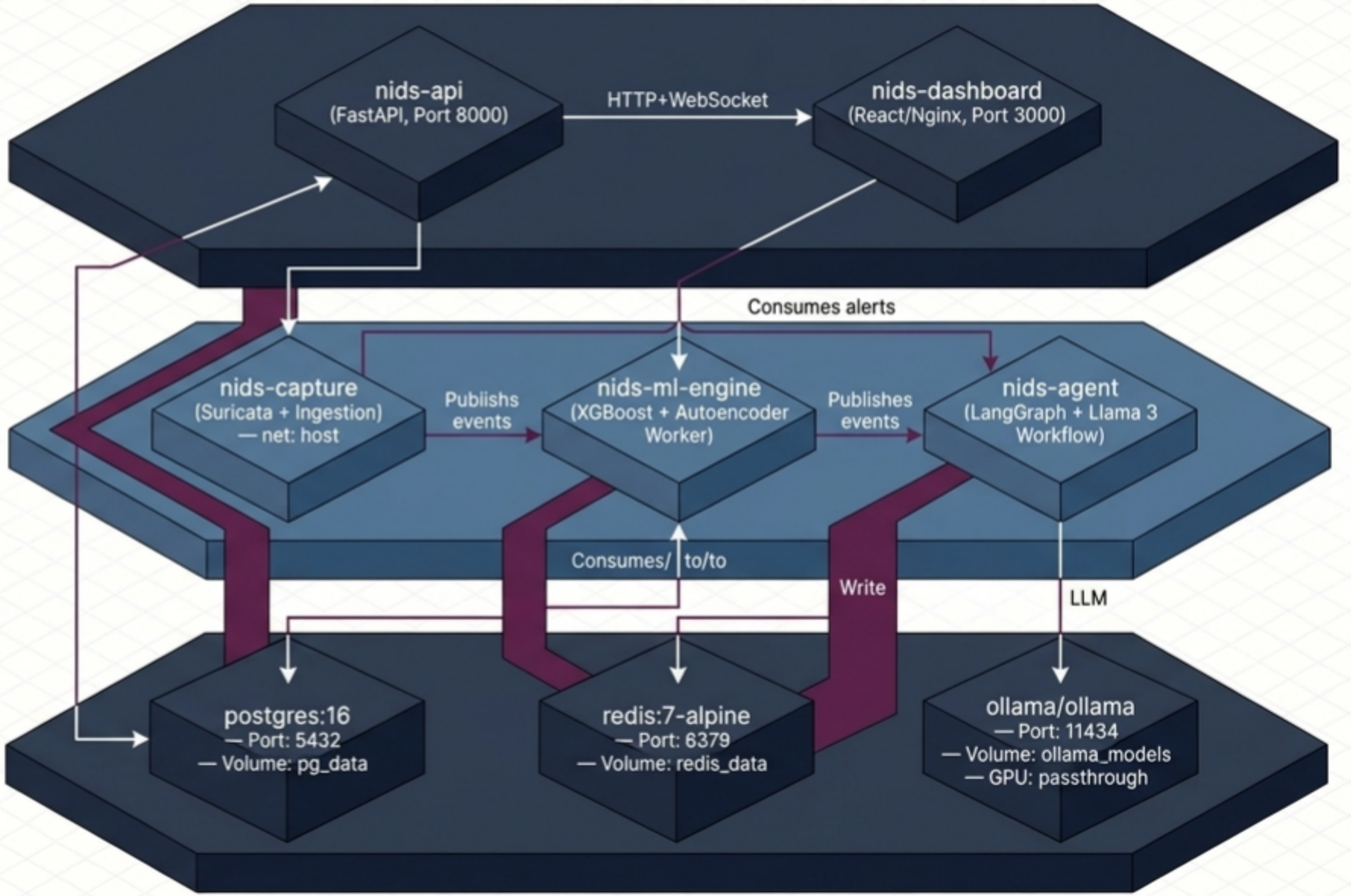
nids-api (FastAPI, Port 8000) and nids-dashboard (React/Nginx, Port 3000). HTTP+WebSocket.

Row 2: Core Services (Middle)

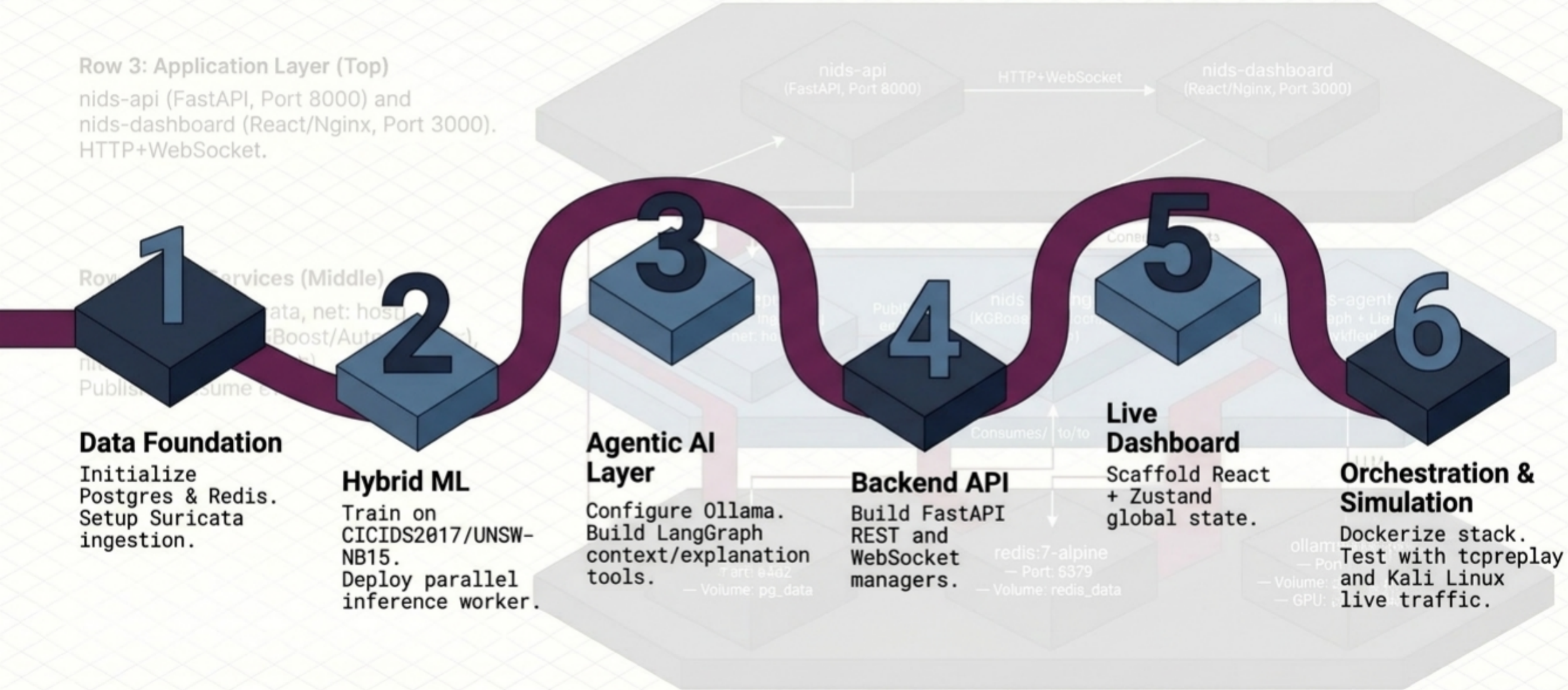
nids-capture (Suricata, net: host), nids-ml-engine (XGBoost/Autoencoder), nids-agent (LangGraph). Publish/Consume events.

Row 1: Infrastructure Services (Bottom)

postgres:16 (Volume: pg_data), redis:7-alpine, ollama/ollama (GPU Passthrough for fast local LLM inference).



The Implementation Roadmap



Data Foundation

Initialize Postgres & Redis. Setup Suricata ingestion.

Hybrid ML

Train on CICIDS2017/UNSW-NB15. Deploy parallel inference worker.

Agentic AI Layer

Configure Ollama. Build LangGraph context/explanation tools.

Backend API

Build FastAPI REST and WebSocket managers.

Live Dashboard

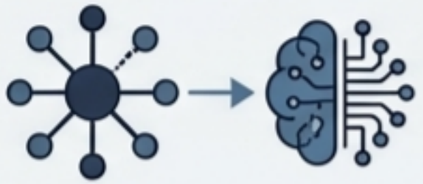
Scaffold React + Zustand global state.

Orchestration & Simulation

Dockerize stack. Test with tcpreplay and Kali Linux live traffic.

Academic & Research Foundation

Validating the architecture against 2022+ peer-reviewed literature.



1. Deep Learning NIDS (2022-2023)

Validates the shift to Transformer/LSTM-based anomaly detection for zero-day threats over traditional signature engines.



2. LLMs in Cybersecurity (2024)

Highlights new opportunities for LLMs in incident response planning and evidence summarization.



3. Multi-Agent Collaboration (2024)

Proves the efficacy of distinct agent personas (e.g., correlation vs. reporting) resolving complex security workflows.



4. Agentic AI in SOCs (2025)

The cutting edge: hyper-automation architectures optimizing SOC KPIs via local autonomous agents.

Demonstrating the Final Project: Simulation & Interaction

1. Live Traffic Simulation & Detection

Replay historical PCAP files (e.g., CICIDS2017, UNSW-NB15) using tools like `tcpreplay` or generate live traffic from Kali Linux. Showcase real-time detection by the Suricata and Hybrid ML engine on the dashboard.

2. Agentic AI Reasoning & Investigation

Trigger a complex alert to demonstrate the multi-agent system's workflow. Show Show how agents collaborate to correlate data, perform context-aware analysis, and provide natural language explanations for the security event.

3. Interactive Dashboard & Workflow

Navigate the React-based dashboard to view live alerts, explore the AI's investigation results, and interact with the system from an analyst's perspective, highlighting the streamlined decision-making process.